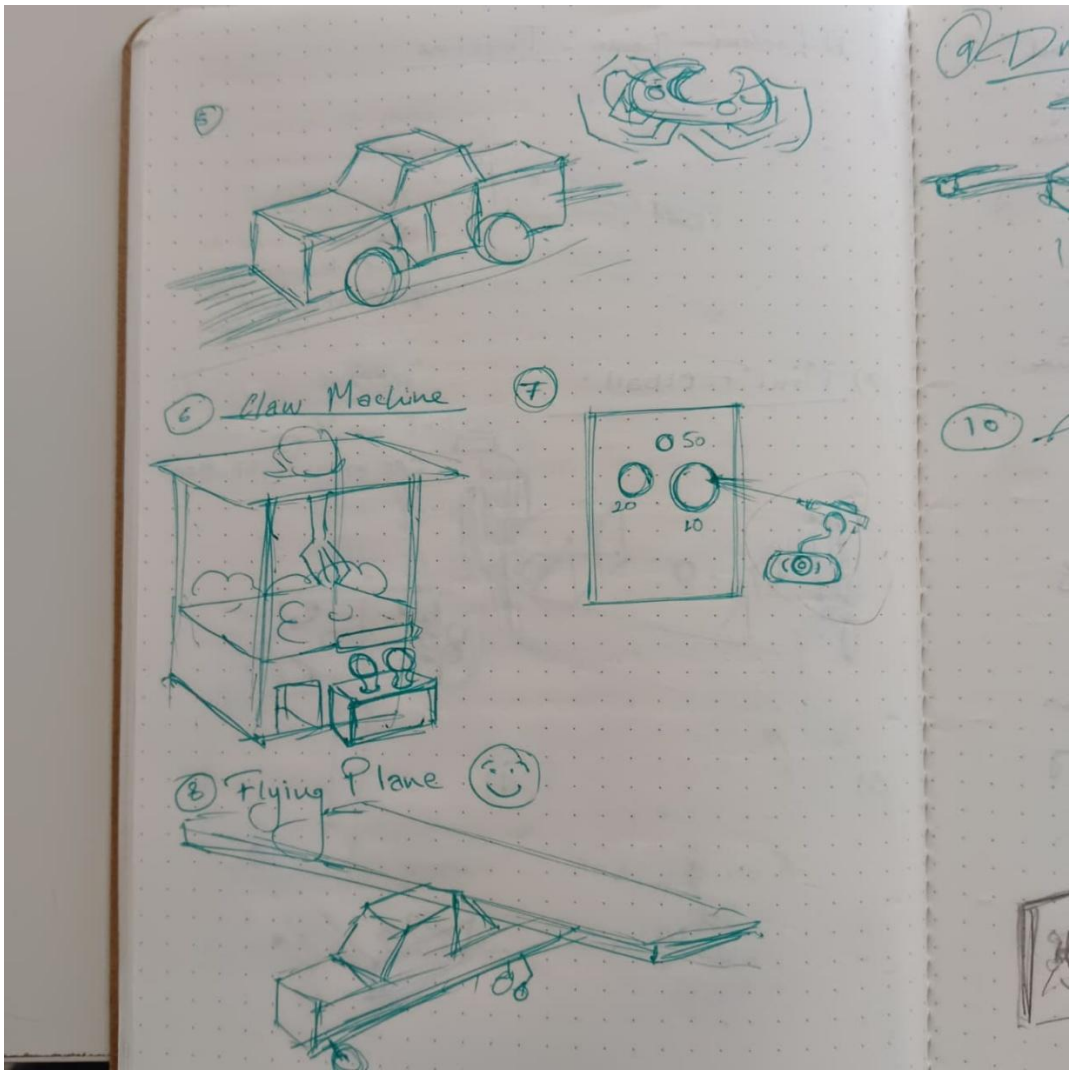
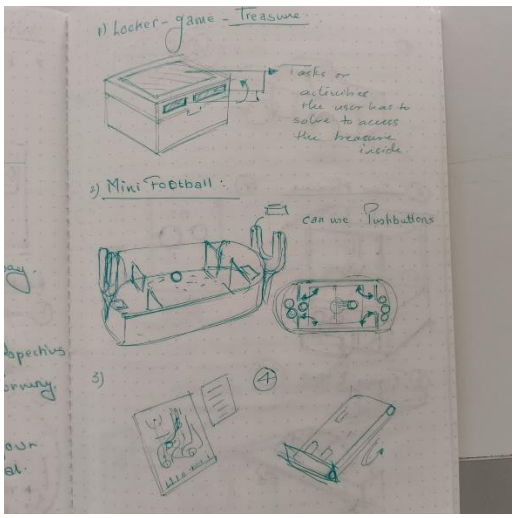
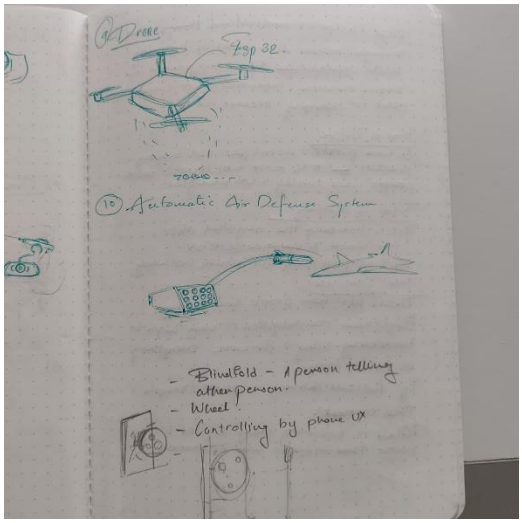
INITIAL IDEAS:







We landed with this idea “Spinshot”, after combining our interests such as the APS turret system, the shooting arcades, and having a physical shooting mechanism, which tested our mechanical skills, and felt more engaging than having a simple laser system.

We chose MDF as our main target board’s body due to its stability and durability for such a model. But for the crossbow stand, we decided to use foam board due to its flexibility and form-changing factor. So, making adjustments was easier and less time-consuming, as the foam board could be carved in any way needed.

INTEGRATION:

Due to the precise model built on LibreCAD, the MDF boards fit perfectly, interlocking with one another. While the foam board provided little difficulty in movement, due to its surface having little friction. Plus, it couldn’t withstand the weight of the motors properly, resulting in a slant focus area for the gun, which can affect the shooting mechanism.

The integration of the breadboard into the MDF box, and the jungle of wires made it difficult to track down each component, making it harder to place or remove. Sometimes, a few jumper wires were seen detached, but it took us some time to figure out where they actually belonged. We kind of solved this problem by taping together jumper wires connected to the same component.